

TIN TIMOTHY KO

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EDUCATION

California State University, Fullerton. GPA: 3.87/4.00 <i>Master of Science in Computer Science</i>	Aug. 2024 - May 2026 Fullerton, CA
National Central University. GPA: 3.67/4.00 <i>Bachelor of Science in Computer Science and Information Engineering (CSIE)</i>	Sep. 2019 - Jun. 2023 Taoyuan, Taiwan

PROFESSIONAL EXPERIENCE

Software Engineer (Part-Time) <i>Sweep Digital Inc.</i>	Jul. 2025 - Present Remote, USA
- Engineered a cross-platform analytics platform in React, React Native, and Supabase, bridging a browser extension and a mobile app to deliver behavioral insights to users. - Architected an event-driven RAG system on Supabase, GCP, and React, transforming 27,000+ browsing events into structured insights for a 50-user beta group. - Built a Python and GCP serverless backend layer with a SetFit model that compressed input data over 80%, cutting GPT-5 API costs.	
AI Automation Specialist Intern (Winter) <i>Unis LLC</i>	Jan. 2025 - Feb. 2025 Buena Park, CA
- Deployed a Python (Flask) and React full-stack driver self-check-in system, integrating with the in-house Warehouse Management System to reduce gate wait times by 70%. - Developed a React, TypeScript, and CSS batch QR generator to streamline inventory tagging workflows for warehouse operators.	
Co-Founder & Lead Engineer <i>Eumius Inc.</i>	Jul. 2023 - Aug. 2024 Los Angeles, CA
- Architected a React Native and GraphQL HealthTech MVP from concept to high-fidelity prototype, collaborating directly with stakeholders to meet market requirements. - Integrated Firebase to build 80% of the MVP, including authentication and real-time medical synchronization for users.	
AI Research Intern <i>Hon Hai Research Institute (Foxconn)</i>	Mar. 2022 - Jun. 2023 Taipei, Taiwan
- Optimized image pre-processing for large ImageNet datasets with Python multi-threading, reducing processing time by 90%. - Built automated Python model-benchmarking pipelines, accelerating weekly experiment validation by 90%. - Applied prompt engineering and reinforcement learning techniques to enhance the OCEAN personality traits of LLMs by 23%.	

PROJECTS

Tech Trend Tracker <i>Go, Python, PostgreSQL (pgvector), Gemma4, RabbitMQ, React, Tailwind CSS, GCP</i>
- Built a Python, RabbitMQ, and Scrappy pipeline to ingest thousands of job postings, decoupling scraping from AI extraction for independent scaling and fault tolerance. - Engineered a local Gemma4 and Ollama NLP engine to extract and normalize the technical skills demanded by each posting into clean, deduplicated data, enabling accurate trend analysis while eliminating 100% of recurring LLM API costs. - Leveraged Go, PostgreSQL, React, and Tailwind on GCP to power sub-second job tech skill searches and project recommendations via hybrid retrieval (pgvector semantic + pg_trgm lexical).

GPU Accelerated Access Control Research <i>C/C++, CUDA</i>
- Engineered custom C++ and CUDA kernels, decreasing memory usage 32x during ARBAC reachability analysis. - Implemented a hybrid GPU/CPU architecture for reachability analysis, achieving a 1400x speedup compared to baseline CPU version. - Built a backward-slicing preprocessor that pruned up to 85% of irrelevant policy rules from the GPU workload.

CompanyWizard <i>React, TypeScript, Next.js, CSS, FastAPI, FastMCP, SQLite, Python, ChromeDB</i>
- Built a Next.js & FastAPI enterprise AI assistant for querying internal systems, reducing support bottlenecks. - Engineered an OpenAI, FastMCP, and ChromaDB RAG pipeline to query live ERP databases. - Built a human-in-the-loop escalation workflow using SQLite, SQLAlchemy, and pytest to update the vector DB with resolved answers.

TECHNICAL SKILLS

Languages: Python, Go (Golang), JavaScript, TypeScript, Java, C/C++, SQL, CUDA, HTML, CSS
Frontend & Mobile: React, React Native, Tailwind CSS, Next.js, HTMX
Backend & Database: Rest APIs, Node.js, Flask, FastAPI, PostgreSQL, MongoDB
Cloud & DevOps: Docker, GCP, Linux, Supabase, Firebase, RabbitMQ, pgmq
AI & Data Processing: PyTorch, RAG, Local LLM, MCP, AI Workflows & Skills, Prompt Engineering
Spoken Languages: English (Native), Mandarin (Native), Cantonese (Native)